JOSE MANUEL ROJAS

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I don't guess. I verify.

AI systems you can verify.

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PROFILE

Seventh-semester Systems Engineering student at Universidad Icesi. I build agents, RAG and predictive models, and surround them with tests, guardrails and human review so they never guess. I use AI as a development team: I define the architecture, the contracts and the acceptance criteria, and the agents work inside their lane.

Looking for an internship and applied AI projects.

METHOD

HOW I WORK WITH AI

- **Contract first, code second.** Requirements and acceptance criteria are written before the first line.
- **Dangerous means blocked.** Push to main, force-push, mass deletes and reading .env are blocked, not politely requested.
- **Two failures and a person steps in.** After two failures the agent investigates; on the third it hands off to a human. Nothing ships without my OK.
- **Every agent in its lane.** A guard blocks edits outside the agent paths, including in parallel branches.
- **Tested or it is not done.** Tests, types, lint and secrets before anything counts as done. Then parallel review.
- **Memory in layers.** Short term: session events. Medium: status pinned in the issue. Long: ADRs and lessons.

PROJECTS

Multimodal RAG: retrieval and fusion across modalities

0.89

RECALL@1 TEXT TO IMAGE

RESEARCH

Artificial Intelligence course, 2026

An experimental study of how to retrieve and combine evidence from text, images, audio and tables in a RAG system, reporting results and limits without gloss.

- Text to image with CLIP ViT-B/32 on Flickr30k: Recall@1 of 0.89 and MRR of 0.936 over 100 queries.
- Text to audio with CLAP on AudioCaps: Recall@10 of 0.98.
- Tables: adding structure (company, region, frequency) raised Recall@1 from 0.67 to 1.00 compared with TF-IDF on a controlled corpus.

The benchmarks are small and partly synthetic, and the work says so.

Python, sentence-transformers, CLAP, scikit-learn, Colab

Winback: reactivation prediction and WhatsApp chatbot

0.82

ROC-AUC ON TEST

IN DEVELOPMENT

AI challenge with a healthcare company

A model that estimates the probability that a former member returns within 12 months, to decide whom to contact first, plus a chatbot that talks and hands off to an advisor anyone who wants to come back.

- 12 models compared with bootstrap confidence intervals. I chose XGBoost for simplicity, because the ensembles tied within the error.
- ROC-AUC of 0.817 and PR-AUC of 0.408 (3.4 times chance). At the recommended threshold it captures 72.8% of reactivations by contacting 28.2%.
- Chatbot with rules for opt-out, complaint and advisor requests without spending LLM calls. The model answers the rest with text RAG and never sees the probability or health data. Sending is simulated, not yet tested with real WhatsApp.

Python, XGBoost, scikit-learn, FastAPI, Next.js, Supabase, pgvector

Rytm: hybrid lifting and running coach

233,280

PROFILES VERIFIED

DEPLOYED PROTOTYPE

Mobile training app (PWA)

An iPhone-installable PWA that guides the lifting session set by set, times cardio, builds a tailored plan and adds an AI coach and a social feed.

- Multi-provider LLM coach (Llama 3.3 on Workers AI by default; Gemini, Groq or Claude optional) that answers in JSON blocks validated by code.
- Deterministic plan engine, no AI: 39 lifting skeletons, 12 cardio templates and 700 tagged exercises, with invariants verified over 233,280 profile combinations.

No users yet.

TypeScript, React 19, Vite, Cloudflare Workers, D1, Durable Objects, Workers AI



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AGENTS WITH THEIR OWN LANE

COMING SOON

With Daniel Escobar

Shift OS and Panel: AI agent teams with human approval

A system for organizing work teams that use AI: 15 agents with their own lane, versioned rules and a panel that shows in real time what each one is doing and lets you approve from the web or over WhatsApp.

- Versioned Claude Code plugin with 15 agents, 8 skills, hooks and a working constitution.
- Real-time panel with Astro, Supabase Realtime and Cloudflare Workers, in production since November 5, 2026.
- My part: co-development, PR reviews and the idea of taking it to work teams as a product, which is the next step.

`Claude Code, Python, Astro, React, Supabase, Cloudflare Workers, n8n`



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TESTS

NOT TESTED LIVE YET

Shift Digital

WhatsApp agent with Claude: automation kit

A customer-service agent for WhatsApp on n8n and Claude Haiku 4.5, with rules enforced by code and not by the prompt.

- A price guard stops any price outside the list from going out and fails closed on formats like «mil» or «k».
- 154 tests, 10 of 10 flows validated against the n8n catalog and 50 of 50 offline acceptance cases.

`n8n, PostgreSQL, WhatsApp Cloud API, Claude Haiku 4.5, Node.js`



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OF MY OWN COMMITS

ADVANCED PROTOTYPE

Extension of an open source CRM (MIT)

WhatsApp CRM for service businesses

I extended an open source CRM with modules for barbershops, dental clinics and sales through conversation.

- Booking agent as a state machine: the AI never decides availability or creates appointments, and a Postgres EXCLUDE constraint prevents double bookings.
- Reception calendar with a copy to Google Calendar (OAuth with encrypted token), a keypad-menu voice robot and post-appointment follow-ups.

`Next.js 16, React 19, TypeScript, Supabase, WhatsApp Cloud API, Vitest`



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PIECES PRODUCED

IN USE

Video production

shift-fotograma: video by code with verified synthetic voice

An engine that renders vertical reels by drawing every frame with code, operated by AI agents.

- Spanish voice generated locally with Chatterbox: a take is accepted only if Whisper detects Spanish, the text matches and the voice resembles the reference.
- Automatic quality control of the final video: loudness at -14 LUFS, audio peak, frozen frames and text outside the safe zone.

Visuals are drawn with code. AI generates the voice and checks the quality.

`Node.js, Python, GSAP, Playwright, ffmpeg, faster-whisper`

EXPERIENCE

Shift Digital 2026 - PRESENT

Co-founder. A launch-stage agency: websites and AI automation for small businesses in Colombia, the United States and Spain.

- Designed and built aishiftdigital.com, in production, with lead capture, a private inbox and an Instagram bot.
- Own products: WhatsApp CRM, customer-service agent with Claude, Mostrador and Shift OS.
- Content production with in-house video engines.

EDUCATION

Universidad Icesi IN PROGRESS, SEVENTH SEMESTER

Systems Engineering. Artificial Intelligence (challenges with industry mentoring) and Internet Computing III.

STACK

AI AND DATA
Claude (Haiku, Sonnet, Opus), OpenAI-compatible APIs (Groq), Workers AI and Gemini, Hybrid RAG (full text and pgvector), CLIP and CLAP, scikit-learn, XGBoost, LightGBM, Prompts with structured output

AGENTS
Claude Code (hooks, skills, subagents), MCP, n8n, OpenCode

LANGUAGES
TypeScript, Python, SQL, JavaScript

WEB AND BACKEND
React 19, Next.js, Astro, NestJS, FastAPI, Tailwind

DATA AND INFRASTRUCTURE
PostgreSQL and Supabase (RLS), Cloudflare (Workers, D1, R2), Vercel and Netlify, Docker

QUALITY
Vitest, Playwright, pgTAP, GitHub Actions

VIDEO AND AUDIO
GSAP, Remotion, ffmpeg, faster-whisper, Chatterbox

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